

Notes: Buchinsky & Hahn (Econometrica, 1998)

An Alternative Estimator for the Censored Quantile Regression Model

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1 Big picture

- **Question.** How can we estimate a censored quantile regression model when censoring creates bias and the standard censored quantile objective is nonconvex?
- **Core contribution.** The paper introduces a censored quantile regression estimator with a linear-programming representation, called CQR-LP.
- **Main advantage.** The CQR-LP objective is globally convex, so the estimator can be computed by linear programming and does not suffer from the local-minimum problem of Powell’s censored quantile regression estimator.
- **Censoring-point advantage.** The estimator does not require direct use of the censoring value for every observation. This lets the censoring point be an unknown function of covariates.
- **Statistical result.** Under regularity conditions, the estimator is root- n consistent and asymptotically normal.
- **Monte Carlo result.** Simulations show that CQR-LP corrects censoring bias well, even with heavy censoring and moderate samples.
- **Economic/econometric takeaway.** The paper converts the censored quantile problem into a weighted uncensored quantile problem on observations whose conditional uncensoring probability is high enough.

2 Model objects and measurement

2.1 Latent-variable Tobit setup

The latent outcome is

$$Y_i^* = X_i' \beta_0 + \varepsilon_{0i}.$$

The observed outcome is left-censored at y_i^0 :

$$Y_i = \max\{y_i^0, X_i' \beta_0 + \varepsilon_{0i}\}.$$

The quantile restriction is

$$\text{Quant}_\theta(\varepsilon_{0i} \mid X_i) = 0.$$

Interpretation. The target parameter is the coefficient vector β_0 that determines the θ -th conditional quantile of the latent uncensored outcome. The observed outcome is censored, so ordinary quantile regression is biased.

2.2 Uncensoring probability and transformed quantile

Let

$$D_i = 1(Y_i^* > y_i^0), \quad h_0(X_i) = \Pr(D_i = 1 \mid X_i).$$

The key conditional quantile index is

$$\tau_0(X_i) = \frac{h_0(X_i) - (1 - \theta)}{h_0(X_i)}.$$

Interpretation. Conditional on being uncensored and having $h_0(X_i) > 1 - \theta$, $X_i' \beta_0$ is the $\tau_0(X_i)$ -th conditional quantile of observed uncensored Y_i .

2.3 Powell estimator and CQR-LP motivation

Powell's censored quantile estimator solves

$$\min_{\beta} \frac{1}{n} \sum_i \rho_{\theta}(Y_i - \max\{y_i^0, X_i' \beta\}),$$

where $\rho_{\theta}(a) = [\theta - 1(a < 0)]a$.

Interpretation. Powell's estimator is semiparametric but nonconvex. CQR-LP replaces it with a globally convex weighted quantile objective.

2.4 Population CQR-LP objective

$$\beta_0 = \arg \min_{\beta} E [D_i 1\{\tau_0(X_i) > 0\} \{\tau_0(X_i)(Y_i - X_i' \beta)^+ + (1 - \tau_0(X_i))(Y_i - X_i' \beta)^-\}].$$

Interpretation. The estimator uses uncensored observations and reweights positive and negative residuals according to $\tau_0(X_i)$.

2.5 Sample estimator with smoothing and trimming

$$\hat{\beta} = \arg \min_{\beta} \sum_i D_i 1\{Z_i \in \mathcal{Z}\} A(\hat{\tau}(X_i)) \{\hat{\tau}(X_i)(Y_i - X_i' \beta)^+ + (1 - \hat{\tau}(X_i))(Y_i - X_i' \beta)^-\}.$$

Interpretation. Smooth trimming controls observations near the boundary $h_0(X_i) = 1 - \theta$, where the transformed quantile is unstable.

3 Models and theoretical specifications

3.1 Kernel estimation of the censoring probability

If $X = (M, Z)$ with discrete M and continuous Z , estimate

$$\hat{h}^{\ell}(z) = \frac{\hat{a}^{\ell}(z)}{\hat{g}^{\ell}(z)},$$

where

$$\begin{aligned} \hat{a}^{\ell}(z) &= (n\delta^k)^{-1} \sum_j D_j 1\{M_j = m^{\ell}\} K((z - Z_j)/\delta), \\ \hat{g}^{\ell}(z) &= (n\delta^k)^{-1} \sum_j 1\{M_j = m^{\ell}\} K((z - Z_j)/\delta). \end{aligned}$$

Interpretation. This first step estimates $\Pr(D = 1 \mid X)$ and is the source of the estimator's bandwidth and dimensionality concerns.

3.2 Asymptotic normality

Under the paper's regularity conditions,

$$\sqrt{n}(\hat{\beta} - \beta_0) \Rightarrow N(0, \theta(1 - \theta)J^{-1}AJ^{-1}),$$

where

$$\begin{aligned} J &= E[1\{Z_i \in \mathcal{Z}\}A(\tau_0(X_i))f_\varepsilon(0 | X_i)X_iX_i'], \\ A &= E[1\{Z_i \in \mathcal{Z}\}A(\tau_0(X_i))^2X_iX_i']. \end{aligned}$$

Interpretation. The variance has the familiar quantile-regression sandwich form, modified by censoring weights and trimming.

3.3 Covariance estimation

$$\begin{aligned} \hat{A} &= \frac{1}{n} \sum_i 1\{Z_i \in \mathcal{Z}\}A(\hat{\tau}(X_i))^2X_iX_i', \\ \hat{J} &= \frac{1}{nh_n} \sum_i 1\{Z_i \in \mathcal{Z}\}A(\hat{\tau}(X_i))1\{|\hat{\varepsilon}_i| < h_n/2\}X_iX_i'. \end{aligned}$$

Interpretation. The density at zero is estimated by counting residuals close to zero, analogous to standard quantile-regression covariance estimation.

3.4 Monte Carlo design

$$\begin{aligned} Y_i^* &= \alpha_0 + X_i'\beta_0 + \varepsilon_i, & Y_i &= Y_i^*1\{Y_i^* > \phi_0\}, & \phi_0 &= -0.75. \\ \varepsilon_i &= u_iv(X_i), & v(x_i) &= \alpha_0 + \sum_j (\alpha_{j1}x_{ij} + \alpha_{j2}x_{ij}^2). \end{aligned}$$

The simulations use $u_i \sim N(0, 25)$ or a recentered $\chi^2(4)$ distribution, and report 10,000 repetitions for the median quantile.

4 Identification and theoretical design

- The conditional quantile restriction identifies β_0 for the latent outcome.
- Censoring is handled by conditioning on uncensored observations and transforming the target quantile to $\tau_0(X)$.
- Given $\tau_0(X)$, the censored problem becomes a weighted uncensored quantile problem with a convex objective.
- The estimator can handle unknown censoring functions because the sample objective uses D and $\hat{h}(X)$, not the censoring value directly.
- Trimming stabilizes observations with $h_0(X)$ close to $1 - \theta$.
- The main practical limitation is estimating $h_0(X)$ in high-dimensional continuous covariates.

5 Tables

5.1 Table 1: Monte Carlo simulation with three regressors for .50 quantile and known censoring point

Read: Table 1 studies a model with a constant and two i.i.d. normal regressors. The top panel uses $N(0, 25)$ errors and has more than 40% censoring. CQR-LP RMSEs and biases shrink as n rises. Powell's estimator has persistent slope bias; in the normal panel at $n = 600$, its slope mean bias is about -0.47 .

Economic message. The convex CQR-LP estimator corrects censoring bias much better than Powell's estimator in this design, especially for the slope parameter.

MONTE CARLO SIMULATION WITH THREE REGRESSORS FOR .50 QUANTILE AND KNOWN CENSORING POINT (10,000 REPETITIONS)

	Constant					Slope						
	CV	CVa	PR	HO	HOa	CR	CV	CVa	PR	HO	HOa	CR
N(0,25) Distribution												
Obs. (100)	54	54	47	54	53	89						
RMSE	2.88	3.34	1.59	4.39	3.02	4.11	2.16	2.10	2.18	2.41	1.98	2.85
Mean bias	0.14	0.07	0.60	0.18	0.44	-0.08	0.31	0.28	0.70	0.32	0.40	0.33
MAE	0.82	0.80	0.94	0.83	0.92	0.74	0.86	0.83	1.21	0.85	0.90	0.92
Median bias	0.39	0.32	0.70	0.45	0.60	0.35	0.06	0.01	0.49	0.05	0.13	-0.31
Obs. (400)	225	226	205	223	219	388						
RMSE	0.58	0.57	0.56	0.60	0.66	0.68	0.61	0.59	0.90	0.67	0.71	0.66
Mean bias	0.20	0.17	0.17	0.19	0.30	0.19	-0.06	-0.10	0.28	-0.05	0.04	-0.45
MAE	0.39	0.38	0.36	0.41	0.45	0.41	0.39	0.39	0.53	0.43	0.44	0.55
Median bias	0.20	0.16	0.16	0.19	0.29	0.19	-0.12	-0.17	0.21	-0.12	-0.05	-0.52
Obs. (600)	340	342	317	338	333	584						
RMSE	0.48	0.48	0.46	0.48	0.52	0.49	0.50	0.48	0.71	0.54	0.56	0.57
Mean bias	0.18	0.16	0.14	0.12	0.20	0.20	-0.06	-0.11	0.25	-0.17	-0.10	-0.47
MAE	0.33	0.32	0.31	0.33	0.35	0.33	0.33	0.33	0.44	0.40	0.40	0.50
Median bias	0.18	0.16	0.15	0.13	0.20	0.19	-0.10	-0.15	0.23	-0.22	-0.16	-0.49
$\chi^2(4)$ Distribution												
Obs. (100)	66	66	62	66	65	95						
RMSE	0.63	0.62	0.62	0.65	0.70	0.67	0.72	0.70	0.92	0.76	0.79	0.90
Mean bias	0.26	0.23	0.25	0.30	0.37	0.20	-0.05	-0.08	0.19	-0.01	0.04	-0.14
MAE	0.41	0.40	0.39	0.43	0.45	0.39	0.46	0.45	0.54	0.48	0.49	0.56
Median bias	0.24	0.21	0.22	0.27	0.33	0.19	-0.11	-0.13	0.12	-0.08	-0.03	-0.37
Obs. (400)	265	266	256	263	260	384						
RMSE	0.32	0.31	0.28	0.32	0.35	0.31	0.35	0.35	0.39	0.37	0.37	0.51
Mean bias	0.17	0.16	0.09	0.17	0.21	0.12	-0.11	-0.14	0.00	-0.09	-0.07	-0.45
MAE	0.22	0.21	0.18	0.22	0.23	0.20	0.24	0.24	0.25	0.25	0.25	0.48
Median bias	0.17	0.16	0.08	0.17	0.20	0.11	-0.13	-0.15	-0.02	-0.11	-0.08	-0.47
Obs. (600)	399	398	385	394	390	582						
RMSE	0.26	0.26	0.23	0.27	0.29	0.25	0.30	0.30	0.32	0.30	0.30	0.50
Mean bias	0.14	0.14	0.06	0.15	0.18	0.09	-0.10	-0.13	-0.01	-0.09	-0.07	-0.46
MAE	0.18	0.18	0.15	0.18	0.19	0.17	0.21	0.21	0.22	0.21	0.21	0.48
Median bias	0.14	0.13	0.06	0.14	0.17	0.10	-0.12	-0.14	-0.03	-0.10	-0.08	-0.48

Notes: The model includes three regressors: a constant and two random i.i.d. $N(0, 1)$ regressors. The vector of coefficients is $(1, 1, 5)$. The censoring point is set at -75 .
CV denotes the CQR-LP estimator with log likelihood cross-validated bandwidth. CVa denotes the CV estimator with bandwidth adjusted to conform with Assumption K. PR denotes the CQR-LP estimator with probit estimates for the censoring probability. HO and HOa are the same as CV and CVa, respectively, except that the kernel function is a higher order kernel. CR denotes Powell's estimator.

observations, the bias in the slope coefficient based on Powell's estimator is about 47%, while for the CQR-LP it is considerably smaller.

The results for the χ^2 distribution are somewhat better than for the normal distribution for all the CQR-LP alternatives, while for Powell's estimator there is no significant difference. Note that deviations from Assumption K, in terms of

MONTE CARLO SIMULATION WITH SIX REGRESSORS FOR .50 QUANTILE AND KNOWN CENSORING POINT (10,000 REPETITIONS)

	Constant				Slope			
	CV	CVa	PR	CR	CV	CVa	PR	CR
$N(0, 25)$ Distribution								
Obs. (100)	51	51	43	82				
RMSE	3.29	3.25	4.36	3.93	2.24	2.24	2.74	3.05
Mean bias	1.66	1.38	3.59	0.70	0.26	0.21	0.72	0.43
MAE	2.07	1.89	3.56	1.55	1.12	1.09	1.60	1.11
Median bias	1.77	1.54	3.53	1.22	0.05	0.00	0.51	-0.12
Obs. (400)	212	213	171	379				
RMSE	1.34	1.28	1.46	1.31	0.78	0.77	1.15	0.89
Mean bias	0.78	0.62	1.15	0.74	-0.19	-0.24	0.26	-0.38
MAE	0.89	0.80	1.17	0.81	0.52	0.51	0.69	0.66
Median bias	0.79	0.64	1.15	0.78	-0.25	-0.30	0.19	-0.54
Obs. (600)	321	324	263	587				
RMSE	1.11	1.01	1.07	0.91	0.63	0.63	0.88	0.71
Mean bias	0.67	0.54	0.81	0.64	-0.14	-0.16	0.16	-0.50
MAE	0.74	0.66	0.84	0.65	0.45	0.45	0.54	0.63
Median bias	0.67	0.56	0.81	0.62	-0.20	-0.22	0.11	-0.60
$\chi^2(4)$ Distribution								
Obs. (100)	61	61	52	91				
RMSE	1.53	1.46	2.01	1.33	1.02	1.03	1.34	1.23
Mean bias	0.96	0.87	1.63	0.69	-0.22	-0.24	0.10	-0.12
MAE	0.99	0.93	1.54	0.75	0.66	0.64	0.80	0.73
Median bias	0.93	0.84	1.53	0.68	-0.30	-0.34	-0.02	-0.37
Obs. (400)	244	244	214	381				
RMSE	0.88	0.80	0.69	0.62	0.51	0.52	0.57	0.60
Mean bias	0.72	0.62	0.51	0.46	-0.24	-0.29	-0.04	-0.47
MAE	0.70	0.61	0.52	0.46	0.37	0.38	0.38	0.54
Median bias	0.70	0.60	0.50	0.45	-0.27	-0.31	-0.07	-0.52
Obs. (600)	365	367	323	583				
RMSE	0.80	0.72	0.54	0.54	0.42	0.43	0.45	0.57
Mean bias	0.69	0.60	0.40	0.42	-0.17	-0.19	-0.04	-0.50
MAE	0.69	0.60	0.41	0.41	0.29	0.31	0.30	0.53
Median bias	0.69	0.60	0.40	0.41	-0.18	-0.20	-0.06	-0.53

Notes: The model includes six regressors: a constant and five random i.i.d. $N(0, 1)$ regressors. The vector of coefficients is $(1, 1, 5, -1, -5, 25)$. The censoring point is set at -75 .
CV denotes the CQR-LP estimator with log likelihood cross-validated bandwidth. CVa denotes the CV estimator with bandwidth adjusted to conform with Assumption K. PR denotes the CQR-LP estimator with probit estimates for the censoring probability. CR denotes Powell's estimator.

both the kernel function employed and the convergence rate of its bandwidth, have only a small effect on the performance of the CQR-LP estimator.

With these results we turn to the simulation results for six regressors.¹⁸ As is expected, the results are not as good as for three regressors. Nevertheless, the

¹⁸ Since the results for the higher order kernel were very similar to those of the normal kernel, we restrict our attention only to the normal kernel.

5.2 Table 2: Monte Carlo simulation with six regressors for .50 quantile and known censoring point

Read: Table 2 increases the dimension to six regressors. The design is harder, but CQR-LP still improves with sample size. At $n = 600$, Powell's slope bias remains about -0.50 , while CQR-LP versions are much closer to zero.

Economic message. The higher-dimensional experiment confirms that CQR-LP is not only computationally convenient but also better centered in finite samples.

5.3 Additional trimming experiments

Read: The paper varies trimming of Z and the smooth trimming value c . More severe trimming of Z worsens performance somewhat, but changing c from 0.005 to 0.05 or 0.10 has almost no effect.

Economic message. The estimator is reasonably robust to trimming choices, making it more practical for applied work.

6 Short summary

- The paper proposes CQR-LP, a censored quantile regression estimator with a globally convex linear-programming objective.
- The estimator handles fixed known censoring and censoring points that are unknown functions of covariates.
- The central transformation is $\tau_0(X) = [h_0(X) - (1 - \theta)]/h_0(X)$.
- Under regularity assumptions, the estimator is root- n consistent and asymptotically normal.
- Monte Carlo simulations show strong finite-sample performance and better centering than Powell’s estimator.
- The main practical limitation is the cost and dimensionality of estimating the censoring probability.

7 Conclusion

7.1 Core conclusion

Buchinsky and Hahn provide a computational and theoretical improvement for censored quantile regression. CQR-LP keeps the semiparametric appeal of quantile methods while replacing a nonconvex censored objective with a convex weighted quantile problem.

7.2 Economic and econometric intuition

Censoring removes part of the latent distribution. Among observations that are not censored, the latent θ -th quantile corresponds to a different quantile of the observed uncensored distribution. That new quantile depends on $h_0(X)$.

7.3 Why the paper matters

The estimator makes censored quantile regression more computationally reliable, broadens the class of censoring rules that can be handled, and allows applied researchers to study heterogeneous effects under censoring without relying only on mean Tobit models.

7.4 Limitations and implementation cautions

Kernel estimation of $h_0(X)$ can be difficult in high dimension. Bandwidth and trimming choices matter in principle, and the method relies on the conditional quantile restriction and support conditions.

7.5 Takeaway

The lasting lesson is that censoring can be corrected by reweighting uncensored observations at the appropriate covariate-dependent quantile. CQR-LP turns this insight into a convex optimization problem.